

Tahsin Reza

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EDUCATION

Bangladesh University of Engineering and Technology (BUET), Dhaka

Jan 2022 - June 2026

Bachelor of Urban and Regional Planning (BURP)

CGPA: 3.70/4.00

Relevant Coursework: Statistics I and II, Programming Techniques, GIS and Remote Sensing, Transportation Policy and Planning, Environmental Planning and Urban Planning Techniques

Academic Honors and Awards: University Merit Scholarship, BUET (2023)

RESEARCH INTEREST

Travel Behavior & Mode Choice, Traffic Modeling and Micro-simulation, Data Science for Transport Planning, Geospatial Analytics for Transportation, Urban Mobility, Safety, Accessibility & Equity in Transportation

TRB ANNUAL MEETING 2027 (ACCEPTED PAPER)

Pair-Specific Time Headway Distributions and Covariate-Conditioned AFT Models for Pedal Rickshaws and Battery-Operated Three-Wheelers in Dhaka's Non-Lane-Based Heterogeneous Traffic.

- Built an automated extraction pipeline, a YOLO26 detector trained on roughly 207,000 self-labeled vehicle instances paired with BoT-SORT tracking, producing about 900 follower-leader observations across two corridors
- Fitted pair-specific headway distributions and accelerated failure time (AFT) models to explain distributional characteristics and shifts, accounting for covariate effects that feed both simulation modeling and the policy horizon.

RESEARCH EXPERIENCE

(1) Intermodal versus Monomodal Trip-Making in Dhaka: Evidence from a Multilevel Choice Analysis. [1st Author, Manuscript in Preparation]

- Built a four-level (M0–M3) Bayesian multilevel logistic regression and an explainable Random Forest/SHAP pipeline to model monomodal vs. intermodal trip choice across 25,242 trips nested within 3,729 Dhaka households.
- Confirmed intermodality is necessity-driven, not discretionary: households below Tk 40,000/month income and in tin-shed housing (OR $\approx$ 5.9 vs. self-built housing) chained modes most, while most vehicle-ownership categories cut the odds by $\sim$ 90% and higher income by $\sim$ 75%, the reverse of the affluence-linked intermodality reported in European studies.

(2) Following Behaviour and Its Regulation: A Time-Headway Study of Pedal- and Battery-Powered Three-Wheelers in Dhaka's Non-Lane-Based Traffic. [1st Author, Manuscript in Preparation]

- Built a maximum-likelihood location-scale regression of vehicle time headway with an exact kinematic-identity decomposition, applied to 898 video-extracted following events from two Dhaka corridors.
- Battery operated three-wheelers followed 30–39% closer than pedal rickshaws; 66–80% of it a speed effect, not spacing; yielding 2.5 $\times$ higher exposure to critically short following across both sites.

(3) Flow Characteristics of Traditional Rickshaws and Battery-Operated Three-Wheelers in Non-Lane-Based Heterogeneous Traffic of Dhaka. [Bachelor's Thesis; Report Attached]

- Videographic surveys at three Dhaka corridors, manually extracting 531 valid target-leader time-headway observations using a reference-line method.
- Benchmarked six tree-based models and selected a tuned CatBoost model with SHAP analysis identifying speed as the strongest predictor, confirming battery-operated three-wheelers consistently follow with shorter headways than rickshaws - evidence against regulating them as one vehicle class.

SELECTED UNDERGRADUATE PROJECTS

[1] Analyzing Core Rural Road Networks and Land Use Conversion: A Case Study on Selected Upazilas in Bangladesh Rural Planning Studio, Department of Urban and Regional Planning, BUET [Report]

- Surveyed all unions across 14 upazilas via KIIs and GIS buffer analysis (LGED roads, BBS demographics, Sentinel land-use), then built a log-log multiple linear regression model ($R^2=0.44$, $p<0.001$) of core road density. Population ($\beta=0.305$)

and education-institution ($\beta=0.205$) density were strongest predictors. Core roads served ~90% of growth centers/markets, 50–60% of key facilities, amid 81% natural land-use conversion.

[2] A Study on Factors Influencing Access Mode Choice of Metro Rail (MRT) Users in Dhaka City

Social and Physical Survey Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed 345 MRT users at 3 stations to identify first/last-mile access mode determinants. Rickshaw dominated (41%), walking/bus tied at 20%; 84%+ traveled under 5 km and about 77% of trips under 1 km were made on foot. Walking led in Pallabi (shortest distance), rickshaw in Uttara North (lower crime risk), and bus in Agargaon (consistent road width).

[3] Factors Influencing Pedestrian Catchment Area of Major MRT Stations in Dhaka

Transport Planning Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed 717 commuters at 2 stations using random sampling and station-gate video counts, mapping built-environment variables across six GIS buffer rings (250–2000 m) and training a Random Forest Regressor to predict egress-walk distance. Mean distances were 440.8 m and 401.7 m, with 12–17% exceeding the conventional 800 m catchment radius; the model achieved $R^2 = 0.94$, with road density and floor area outweighing sociodemographic factors.

PROFESSIONAL EXPERIENCE

Assistant Urban Planner

July 2026 – September 2026

Sheltech Consultants (Pvt.) Ltd., Dhaka

- Drafted the transportation planning volume of the Pourashava master plan report under IUGIP (Improving Urban Governance & Infrastructure Program), based on the survey report and ToR.
- Refined the LOS determination methodology and supported the wider project team with transport-related queries while reporting for all Pourashavas under the program.

Student Intern

November 2025 - December 2025

Dhaka Transport Coordination Authority (DTCA), Dhaka

- Assessed the traffic impact of converting a road in Dhaka into a pedestrian precinct. Ran traffic and pedestrian surveys, then built and compared before-and-after microsimulation scenarios in PTV Vissim to test the proposed layout.

TEACHING EXPERIENCE

Tutoring Secondary and Higher Secondary students (Physics, Math, ICT, Chemistry)

2021-Present

- Provided academic support to over 12 students across varying levels
- Served as Mathematics instructor at a local coaching center

SOFTWARE AND PROGRAMMING SKILLS

Programming Language: Python, R, C | **GIS Tool:** ArcGIS, ArcGIS Pro, Google Earth Engine | **Transportation Modelling:** PTV Vissim | **Modeling and ML:** Discrete Choice Models, Survival/AFT Models, Tree-Based Ensemble Models, SHAP, YOLO Object Detection | **Design and Drafting:** AutoCAD, Adobe Illustrator, Adobe Photoshop | **Productivity:** MS Word, Excel, PowerPoint, Project, CapCut

EXTRA-CURRICULAR ACTIVITIES

- Secretary, Student Chapter, **Bangladesh Institute of Planners (BIP)** (2025-2026)
- Vice President, **URP Students' Association of BUET (USAB)** (2025-2026)
- Joint General Secretary, **Help_BUETian** (2025-2026), Deputy General Secretary, **Help_BUETian** (2024-2025)

REFERENCES

Dr. Md. Musleh Uddin Hasan

Professor

Dept. of Urban and Regional Planning

Bangladesh University of Engineering and Technology

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Relationship: Thesis Supervisor and Research Mentor

Dr. Md. Asif Raihan

Professor, Accident Research Institute (ARI), BUET

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Relationship: Research Mentor